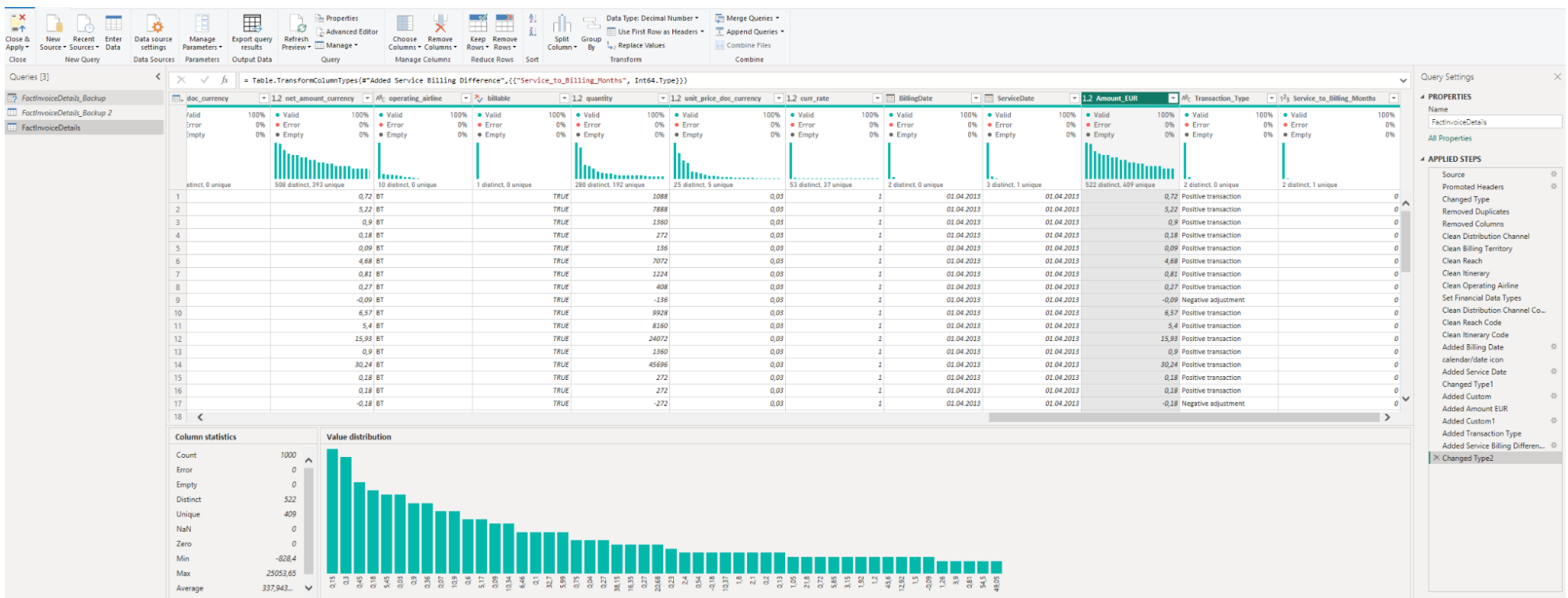


# Power BI Assessment - Short Submission Notes

The notes below summarize what I actually did in the file and the main points I would explain together with the PBIX.

## 1. Assumptions

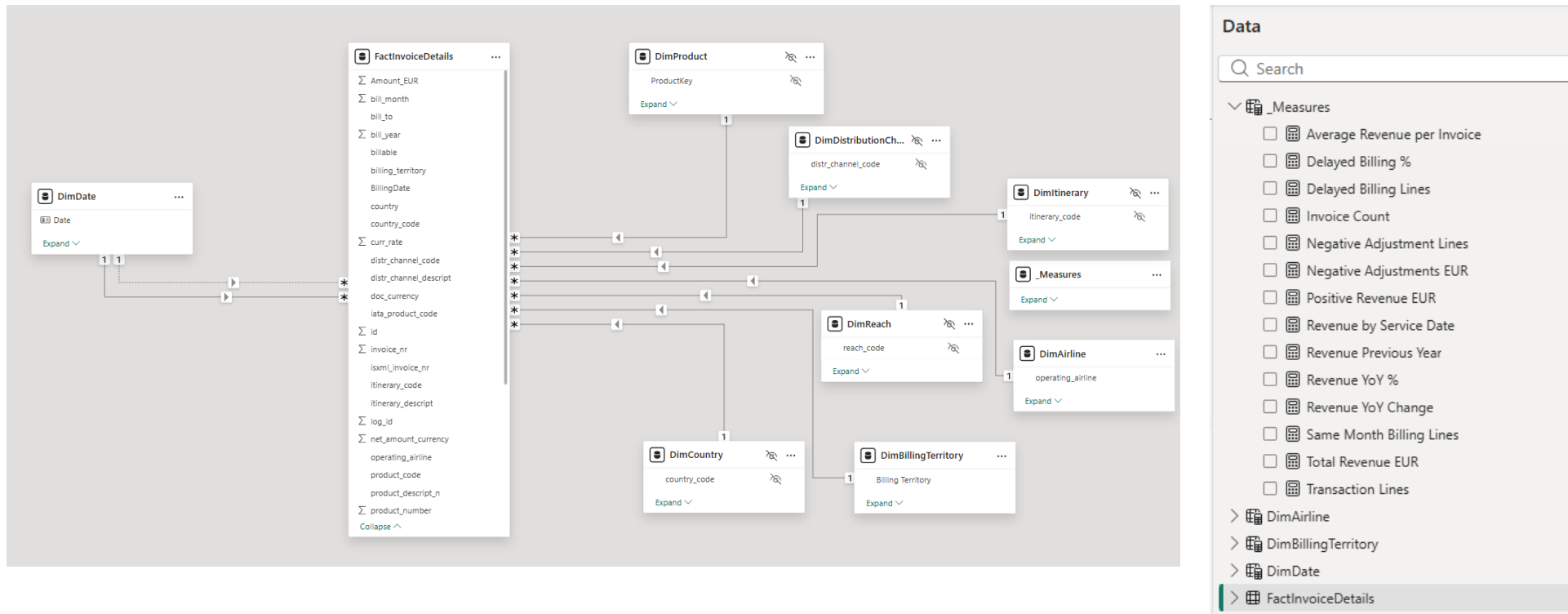
- I used the provided CSV as the source for the task and changed the data only where there was a clear data-quality or reporting reason.
- I removed 640 rows only when the whole row was identical. I did not treat the same invoice number as a duplicate by itself, because one invoice can contain several valid lines.
- I removed 5 completely empty columns and 3 columns that were around 97-99% empty, because they did not add useful information for this amount of data.
- Where a useful descriptive field was blank, I kept the transaction and used 'Unknown' instead of deleting the financial row.
- I kept negative amounts. I treated them as possible adjustments, credits or corrections, because removing them would change the financial result.
- The source contains EUR and USD, so I created one final reporting amount in EUR:  $\text{Amount\_EUR} = \text{net\_amount\_currency} \times \text{curr\_rate}$ .
- The source gives year and month rather than an exact day, so I created BillingDate and ServiceDate using the first day of each month.



Power Query after cleaning, date creation, EUR conversion and data-quality flags.

## 2. Modeling decisions

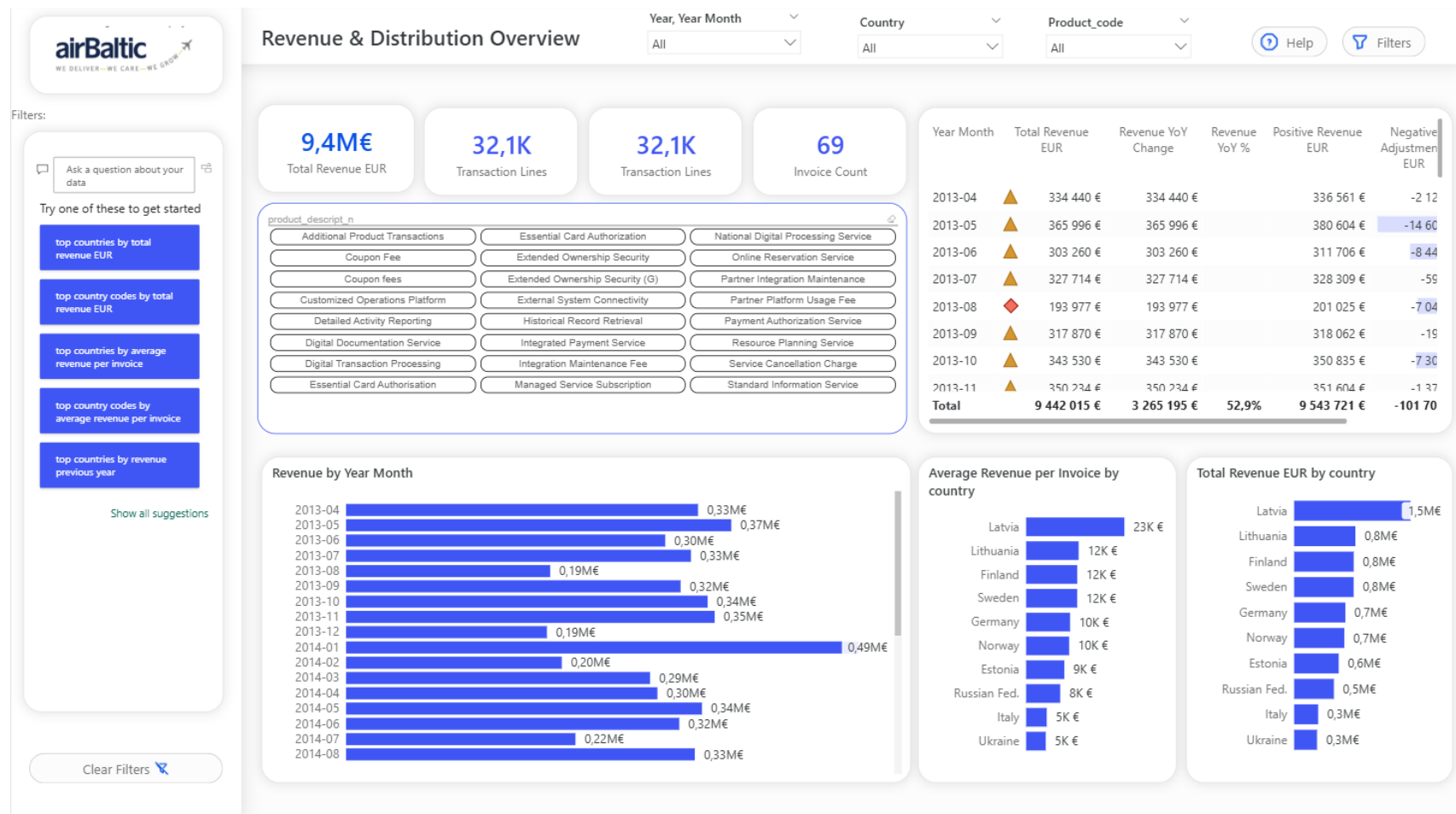
- I cleaned the data in Power Query first and created a backup query before the later transformations. The backup query was kept with load disabled.
- I used a star schema with FactInvoiceDetails in the center and separate dimensions for Date, Product, Country, Distribution Channel, Reach, Itinerary, Airline and Billing Territory.
- Product number was not unique for every product variant. Instead of forcing a many-to-many relationship, I created a composite ProductKey.
- I also separated Country and Billing Territory because combining them caused duplicate country keys.
- The main Date relationship uses BillingDate and is active. ServiceDate is kept as a second inactive relationship so it can still be used when needed.
- Relationships are one-to-many with single-direction filtering. I avoided many-to-many relationships where a cleaner dimension key could be created.
- I kept the main calculations in a separate \_Measures table so the report logic is easier to find and maintain.



Final star schema and the dedicated measures table.

### 3. Key analysis findings

- After cleaning, the dataset contains 32,141 transaction lines and 69 invoices. Total reported revenue is about EUR 9.44M.
- 2014 was the strongest full year at about EUR 3.45M. Revenue in 2015 was about EUR 3.27M, which is around 5.3% lower than 2014.
- 2013 starts in April, so I would not use 2013 vs 2014 as a normal full-year comparison.
- Latvia is the largest market in the dataset at around EUR 1.54M.
- Negative adjustments are around EUR 101.7K. I kept them in the model because they are part of the financial activity.
- Delayed billing is low overall at around 0.5%, but it increases to around 1.0% in 2015.
- Revenue is concentrated in a relatively small number of products and channels. For that reason I added filtering and drill-down by time, country, product and distribution channel so the result can be checked from different angles.



Final executive dashboard used for the submission.